1. Clinical Question & Core Hypothesis

Clinical Question

The Ontologist identified three convergent but sometimes discordant signals in NMIBC — non-invasive urine DNA methylation markers, tumor immune organization (Immunoscore/leukocyte fraction/TLS), and MMR/genomic status (MSH6). Clinically, TLS or high immune infiltration does not uniformly predict benefit from intravesical or systemic immune therapies, and we need to know what molecular features determine when organized local immunity is productive versus dysfunctional. 2) Tier 1 paths used: (a) selected urine–DNA methylation biomarkers — chosen because they provide a non-invasive readout of tumor clonality/epigenetic heterogeneity; (b) immunotherapy biomarker / immune-context path — which motivates using an imaging immune feature (TLS) as the visual correlate of organized local immunity; (c) MSH6 prognostic path — because MSH6-driven MMR changes neoantigen load and can mechanistically modulate immune recognition. These three form a mechanistic axis: antigenicity (MSH6) ↔ epigenetic clonality (urine methylation) ↔ immune organization (TLS). 3) Tier 2 support: Immunoscore/CD3–CD8 metrics and leukocyte fraction justify focusing on organized immune infiltrates; EV mRNA signals (KRT17, MDK) and PD-L1 support the relevance of non-invasive tumor-derived signals and immune-suppressive modifiers; metabolic/ systemic markers (IGF■1, lipids) suggest host-state modifiers to consider later. 4) Why therapy_response as endpoint: Predicting benefit from BCG or immune-based therapies is an urgent, actionable clinical need in NMIBC and is mechanistically tied to both tumor antigenicity (MMR) and local immune organization (TLS). A therapy-response endpoint allows direct translation into patient selection for intravesical or checkpoint strategies. 5) What I excluded and why: I deliberately excluded AI-accessibility/CSF-AD and lymph-node-metastasis paths from the core hypothesis because they are methodological/cross-disease or broad prognostic signals without direct mechanistic linkage to TLS–MSH6–urine methylation interaction; these remain exploratory (Tier 3) or candidates for secondary analyses to preserve hypothesis focus?

Working Hypothesis

In NMIBC, TLS-like lymphoid aggregates predict favorable responses to intravesical BCG or other immunotherapies only when the tumor exhibits MSH6-associated neoantigen enrichment accompanied by coherent, clonal urine DNA methylation signatures. If TLS are present but MSH6-driven neoantigen load is low or urine methylation signatures reflect discordant clonal architecture, TLS do not reliably predict productive anti-tumor immunity and therapy response is diminished. A combined biomarker model integrating TLS organization, MSH6 status, and urine DNA methylation will more accurately stratify NMIBC patients for immune-based therapies and predict therapy_response, recurrence-free survival, and progression-free survival.

Element	Value
Imaging feature	TLS-like lymphoid aggregates
Anchor gene(s)	MSH6, urine-DNA methylation biomarkers
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

MSH6: MSH6–urine methylation interaction; these remain exploratory (Tier 3) or candidates for secondary analyses to preserve hypothesis focus.

urine-DNA methylation biomarkers: Selected as a core mechanistic driver from Tier 1 knowledge graph analysis.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	CD3	Supporting modifier from knowledge graph
Secondary	CD8	Supporting modifier from knowledge graph
Secondary	leukocyte_fraction	Supporting modifier from knowledge graph
Secondary	MDK	Supporting modifier from knowledge graph
Secondary	KRT17	Supporting modifier from knowledge graph
Secondary	PD-L1	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: Use retrospective NMIBC cohorts treated with BCG and/or immunotherapies;

Prediction 2: quantify TLS density/maturity in tumor sections (IHC for CD3, CD8, CD20, follicular dendritic cell markers) and classify TLS;

Prediction 3: assess MSH6 status by sequencing and/or IHC;

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- CD3: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- CD8: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.